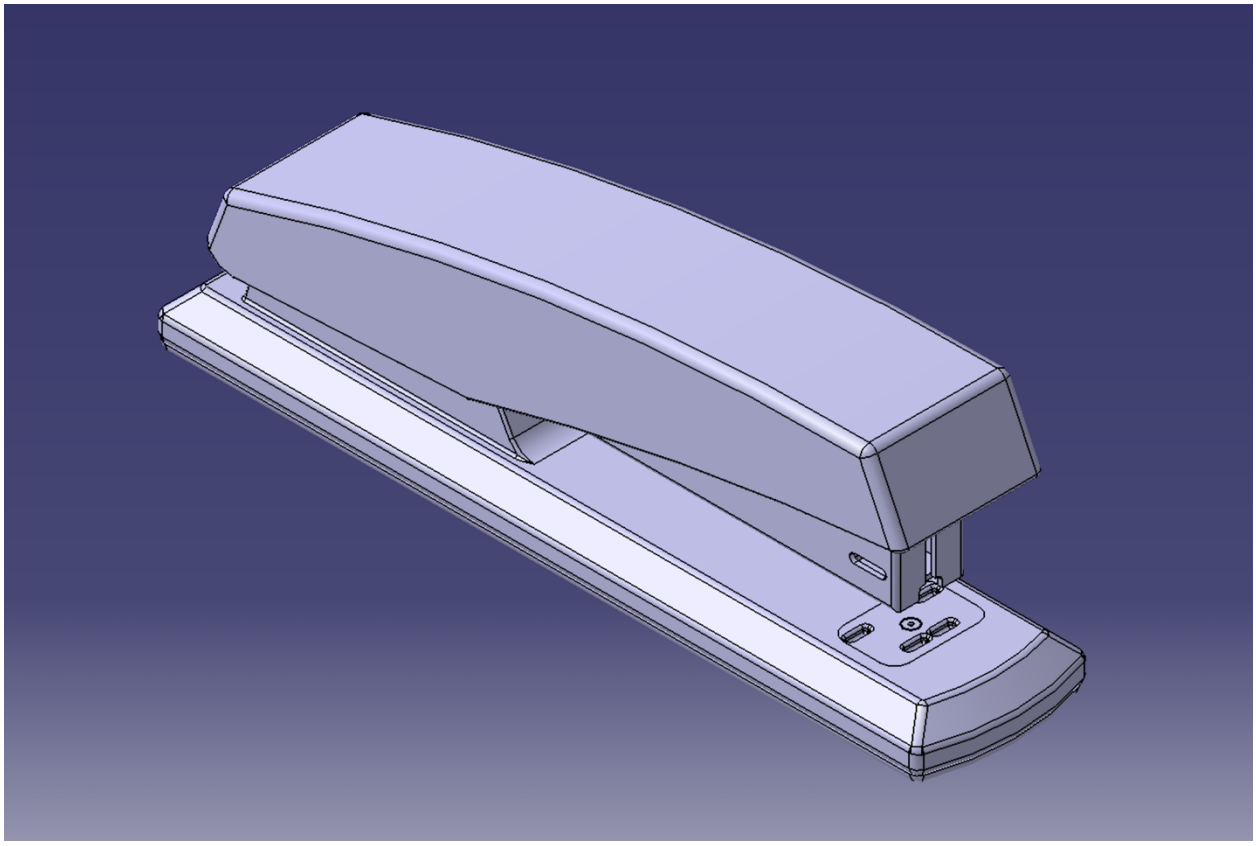
EGR 120 Graphical Communications

CATIA V5 3D Modeling & Drafting



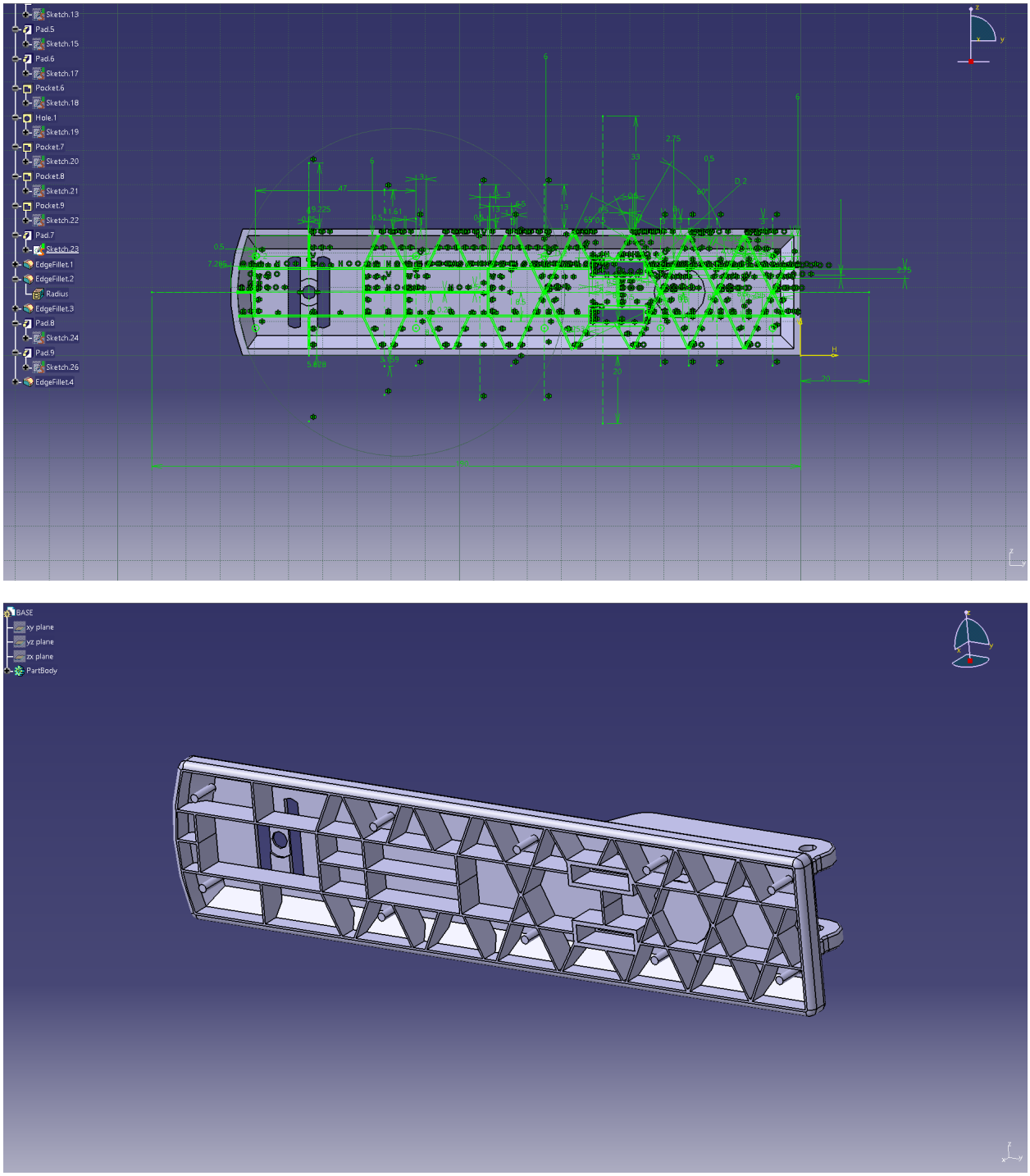
Individual Final Project: Aayin Choi

Project Score: **198**/200

Overall Standing: **Rank #1** Across All Instructor Sections

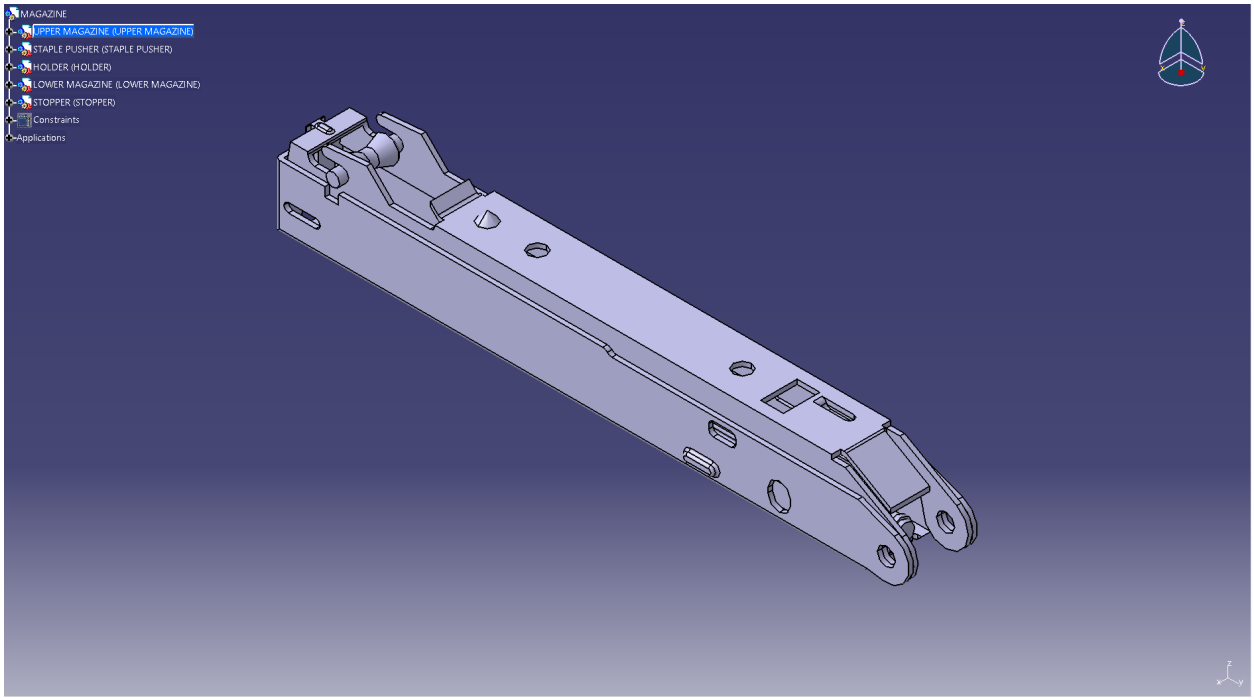
Project Scope: **10** Parts · **2** Subassemblies · **44** Drawing Sheets

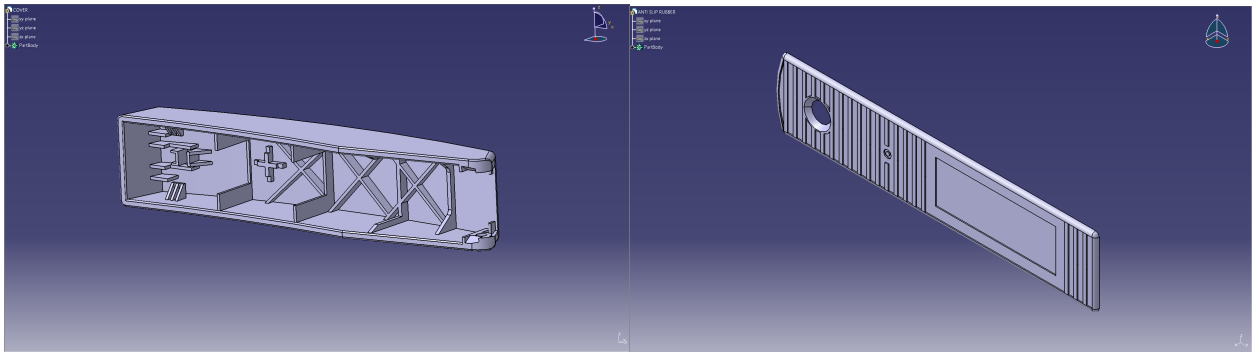
Instructor: Prof. David Richter



These images trace the development of the stapler's base from an extensive network of dimensioned sketches and features to the completed 3D geometry.

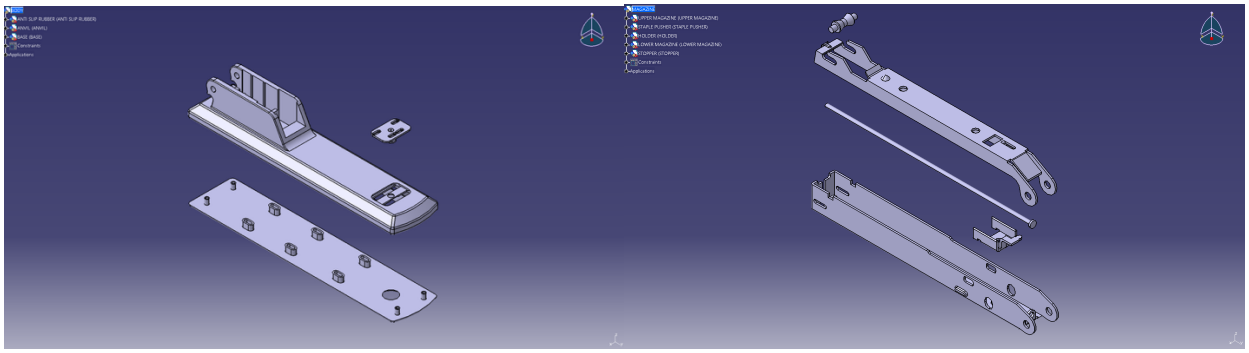
Its numerous internal ribs, cutouts, and mating surfaces made it the most intricate and technically demanding component of the project.

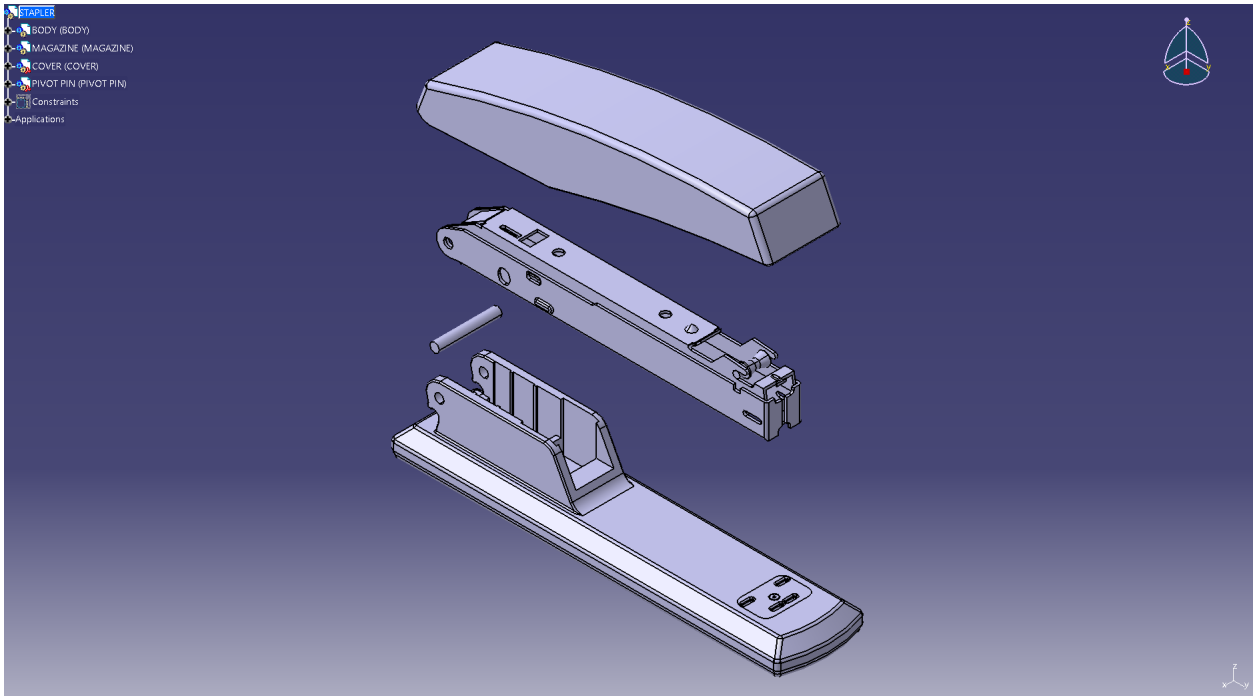




These images present three of the stapler's principal elements: the Magazine, a five-component subassembly consisting of the Upper Magazine, Staple Pusher, Holder, Lower Magazine, and Stopper; the Cover; and the Anti-Slip Rubber.

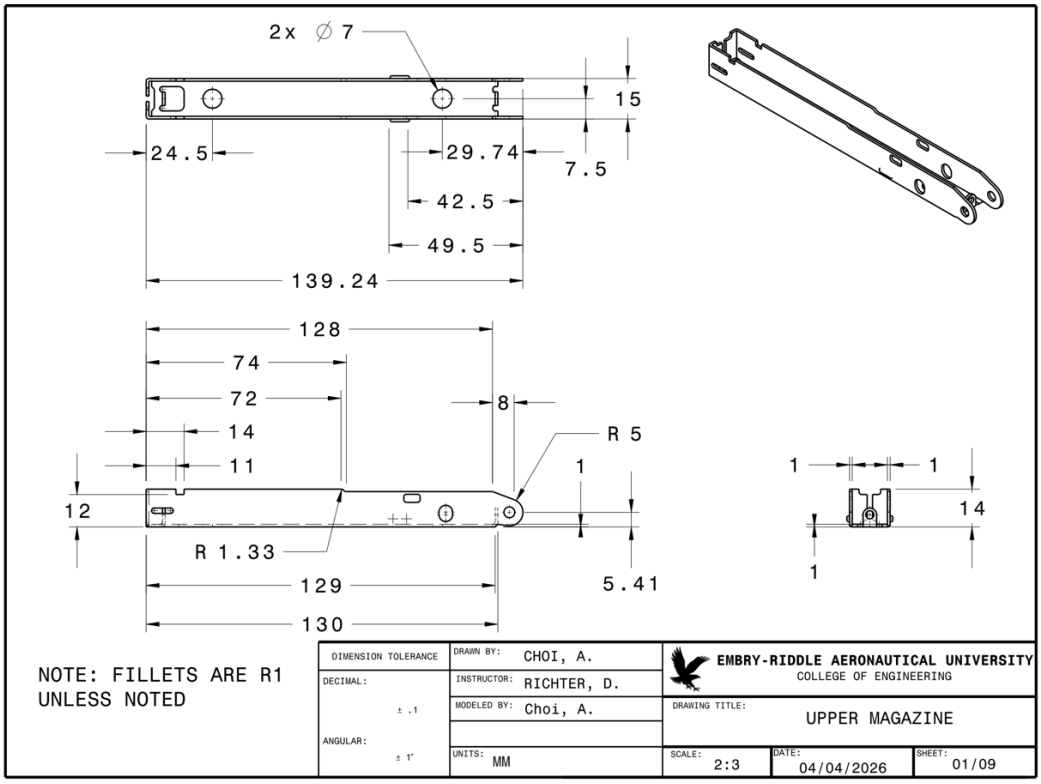
Each element was developed through direct measurement and continuous comparison with the physical stapler, allowing the resulting geometry to closely reproduce the original product's dimensions, proportions, and overall form.

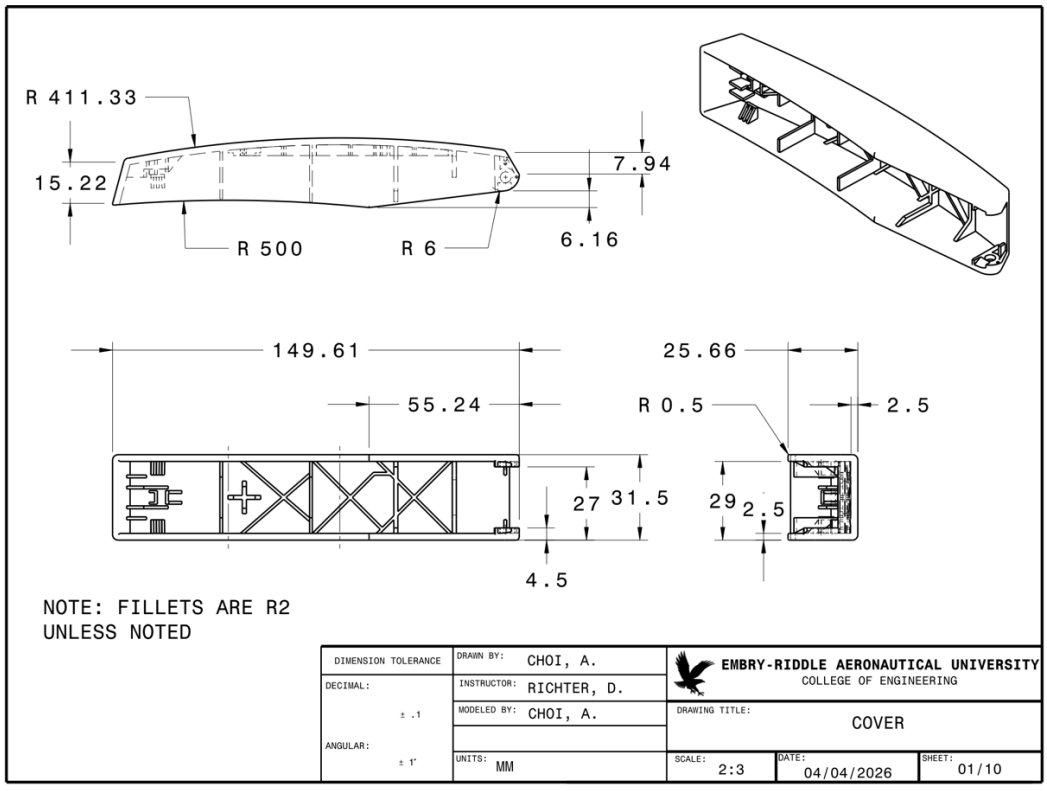




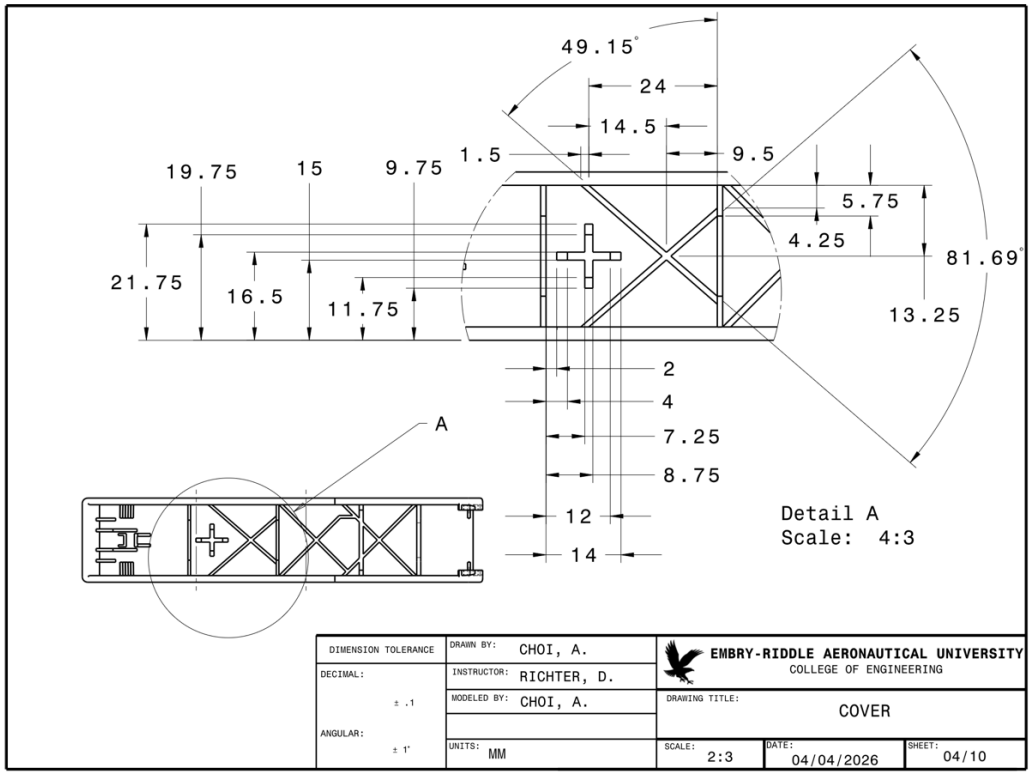
These exploded views document the staged assembly of the stapler from ten individually modeled components. Each component was positioned sequentially using the mating features and interfaces established during development.

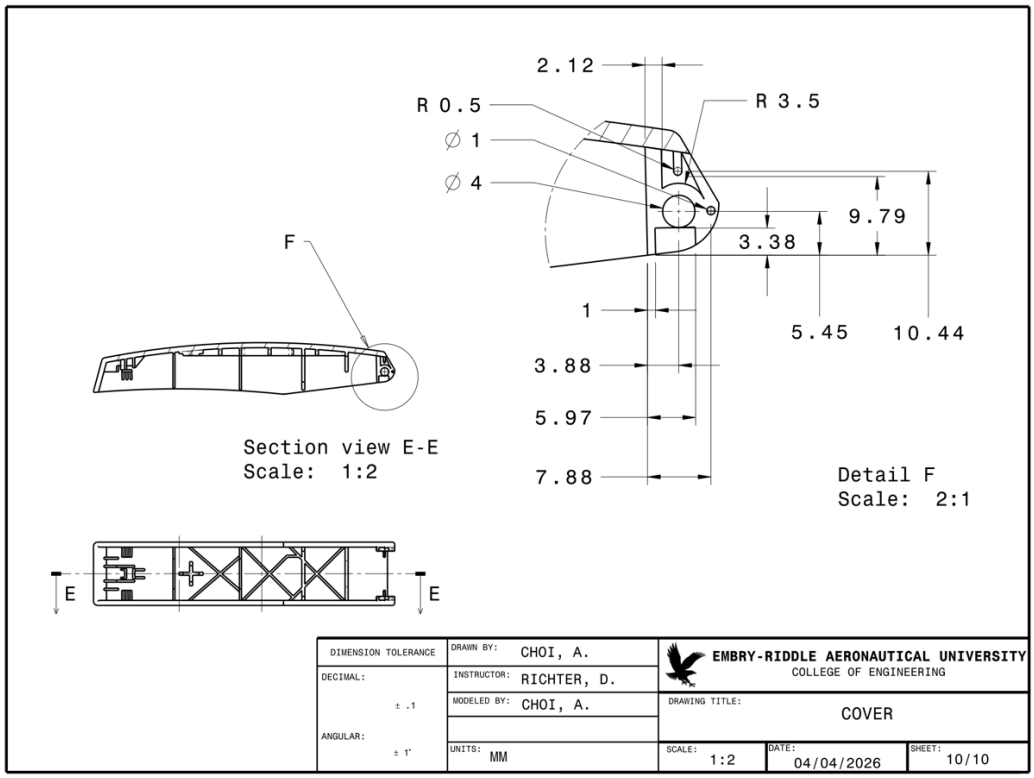
The assembly was completed with minimal dimensional adjustment, and all ten components aligned as intended—validating the dimensional consistency of the individual models. The completed assembly closely reproduces the physical stapler's overall scale, proportions, and component relationships.





Following completion and assembly verification of all ten modeled components, a total of 44 technical drawings were produced to document the finalized design.





The finalized 3D geometry was translated into fully dimensioned orthographic, sectional, detail, and isometric views, clearly documenting each component's dimensions, internal geometry, and critical design features.